YUVRAJ M. SINGH

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EDUCATION

Lokmanya Tilak College of Engineering, Navi Mumbai, India — *May 2025*
Bachelor of Engineering — Electronics & Telecommunication — *CGPA: 7.49*

Laxman Devram Sonawane College, Mumbai, India — *May 2021*
HSC — Science — *Score: 86%*

SKILLS

Languages: Python, Bash, SQL (Oracle 19c / MySQL), JavaScript, TypeScript, HTML/CSS, R

Database & DBA: Oracle 19c, RMAN, OEM (Oracle Enterprise Manager), PL/SQL, MySQL, Schema Administration, Backup & Recovery

Cloud & DevOps: Microsoft Azure (AZ-900, AI-900 certified), AWS, Docker, Git, GitHub, Linux, Windows Server

Data & Libraries: Pandas, NumPy, Matplotlib, Flask, scikit-learn, Power BI

Hardware / Embedded: Arduino, ESP32, ESP8266, HC-05, MSP430, OpenCV

EXPERIENCE

Oracle DBA | **Reliance Jio** *(via Quess Corp)* — Navi Mumbai, India — Sep 2025 – Present

- **Automated Restoration:** Developed a production-grade Bash scripting framework for RMAN-based full & incremental database restoration, featuring error handling, lock-file mechanisms, CLI flag interfaces, and SQL/RMAN wrapper functions — reducing restore time by **85%** and eliminating manual intervention entirely.
- **Database Monitoring:** Integrated 15+ Oracle databases into Oracle Enterprise Manager (OEM) for unified monitoring; configured rule-based alert thresholds for critical events including tablespace usage, session spikes, and redo log issues.
- **Patch Automation:** Developed shell-based SBP patching pipelines for Oracle 19c with pre/post-patch validation, rollback support, and audit logging — standardizing patch cycles across production environments.

Data Analyst Intern | **Optimized Tech Solutions** — Pune, India (Hybrid) — Aug 2024 – Feb 2025

- **Machine Learning:** Trained ensemble ML models (Random Forest, MLP, DNN) in Python to fully automate a manual data parsing workflow; achieved **>96% accuracy** using hyperparameter tuning with grid search and k-fold cross-validation, cutting processing time by **88%**.
- **Data Engineering:** Transformed and cleansed large volumes of unstructured business data using Pandas and NumPy, aligning datasets to business schema requirements and enabling downstream model training pipelines.
- **Product Ideation:** Designed and presented a Q&A upvoting dashboard concept to the CEO to improve internal knowledge-sharing — projected to increase employee engagement by up to **8**×.

PROJECTS (Full portfolio at YuvrajSingh.me)

Smart Home Automation System | *ESP32, Arduino, Blynk IoT, OpenCV, HC-05* — Mar 2023 – Apr 2024

- Designed an extensible IoT home automation system allowing new devices to be added without modifying existing code; supports seamless online/offline fallback with a remote web dashboard for global control.
- Integrated face detection and child-safety features using OpenCV + ESP32-Cam, and replaced conventional switchboards with an IoT smart switch system with automation triggers and remote operation.

Multilingual Speech & Video Translation System | *Python, Flask, WebRTC, Google APIs, gTTS* — May 2024 – Present

- Built a full-stack real-time speech translation platform using WebSockets/WebRTC for low-latency speech-to-text, Google Translate, and Web Speech API for live TTS playback across multiple languages simultaneously.
- Extended with an offline video translation pipeline: video is transcribed, translated, and re-dubbed with synchronized subtitles and generated voiceovers using gTTS — with zero third-party SaaS dependency.

Modular Voice-Controlled Desktop Assistant | *Python, SpeechRecognition, pyautogui, tkinter* — 2024 – Present

- Engineered a modular voice assistant with fully separated components (`speech_engine`, `ai_rewrite`, `whatsapp_handler`, `overlay_gui`, `confirmation`) that interprets voice commands via an AI rewrite layer and executes real desktop actions — currently controlling WhatsApp Desktop for messaging, voice, and video calls.
- Built a floating GUI overlay with live status feedback (*Listening* → *Recognizing* → *Sending*) and a voice-driven confirmation gate before sensitive actions to prevent accidental execution.

CERTIFICATIONS

Azure AI Fundamentals (AI-900) — Ambassador Challenge | *Microsoft* — Jun 2024
Azure Fundamentals (AZ-900) | *Koenig Solutions / Microsoft* — Feb 2024
Azure Cloud Ambassador Challenge | *Microsoft Learning* — May 2024
Project Management Foundations: Teams | *LinkedIn Learning* — Dec 2023